

PRELIM

Introduction to Information Assurance and Security

Course Code : IAS201

Course Title : INFORMATION ASSURANCE AND SECURITY 1

Course Description :

This course covers the examination of the relationship between threats, vulnerabilities, countermeasures, attacks, compromises and remediation throughout the entire system life cycle. It includes explanation of the key factors involved in authentication and how they are used to verify identity and grant access to the system. It also involves description of the legal and ethical consideration related to the handling and management of enterprise information assets.

Week 1 – Introduction to Information Assurance

In today's digital world, organizations heavily depend on information systems to support daily operations, communication, financial transactions, healthcare, education, and government services. As a result, protecting information has become one of the most important responsibilities of IT professionals.

This lesson introduces the fundamental concepts of **Information Assurance (IA)**, **Information Security**, and **Cybersecurity**. Students will learn how these disciplines are related, why organizations invest in them, and why information is considered one of the most valuable organizational assets.

By the end of this lesson, students should understand that cybersecurity is not just about stopping hackers—it is about ensuring that information remains **accurate, available, confidential, and trustworthy** throughout its lifecycle.

Learning Outcomes

At the end of this lesson, students should be able to:

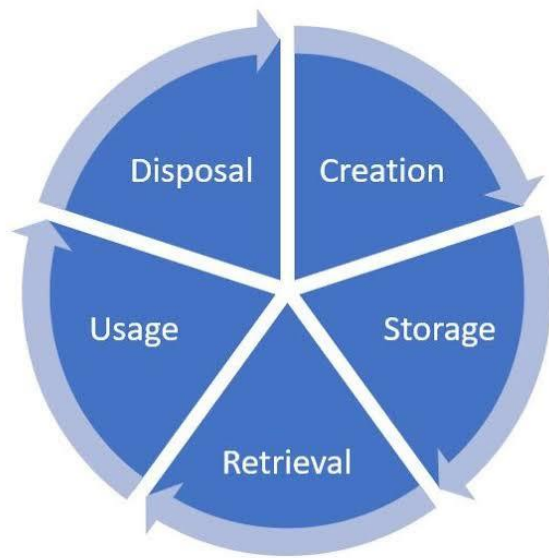
- Define Information Assurance (IA).
- Explain Information Security and Cybersecurity.
- Differentiate IA, Information Security, and Cybersecurity.
- Describe the goals of Information Assurance.
- Explain why information is considered an organizational asset.
- Analyze real-world cybersecurity incidents using IA concepts.

1. What is Information Assurance (IA)?

Information Assurance (IA) is the discipline of protecting information and information systems by ensuring that data remains:

- Confidential
- Accurate
- Available
- Authentic
- Reliable

It combines **people, processes, policies, and technology** to protect information throughout its entire lifecycle—from creation and storage to sharing and secure disposal.



Unlike cybersecurity, Information Assurance also includes **risk management, business continuity, policy development, governance, and legal compliance.**

Key Objectives

Information Assurance aims to:

- Protect sensitive information.
- Ensure business continuity.
- Minimize risks.
- Prevent unauthorized access.
- Maintain customer trust.
- Comply with laws and regulations.
- Ensure the reliability of information systems.

Data Life-Cycle Management <https://www.crownrms.com/uk/insights/information-lifecycle-management/>

Real-World Example: Banking Systems

Imagine you transfer **₱10,000** from your BPI account to another bank.

Information Assurance ensures that:

- Your account is accessed only by you.
- The amount transferred is exactly ₱10,000.
- The banking system is available during the transaction.
- The transaction is properly recorded.
- There is evidence proving the transaction occurred.

Without Information Assurance, your funds could be altered, intercepted, or lost.

Analysis

Many people associate security solely with hackers. However, Information Assurance encompasses much more. A secure organization also needs backup procedures, disaster recovery plans, employee training, and policies that define how information should be handled.

For example, even if a company's firewall is strong, poor employee practices—such as sharing passwords—can still lead to a security breach. This demonstrates that Information Assurance addresses both technical and human factors.

2. What is Cybersecurity?

Cybersecurity is the practice of protecting digital systems, networks, software, and electronic information from cyber threats and attacks.

Cybersecurity focuses on defending against:

- **Hackers** - Individuals who attempt to gain unauthorized access to computer systems or networks.
- **Malware** - Malicious software designed to damage, disrupt, or gain unauthorized access to a computer system.
- **Viruses** - A type of malware that attaches itself to files or programs and spreads when they are executed.



- **Ransomware** - Malware that encrypts a victim's files and demands payment to restore access.
- **Phishing attacks** - Fraudulent attempts to trick users into revealing sensitive information through fake emails, messages, or websites.
- **Data breaches** - Security incidents in which unauthorized individuals access, steal, or expose confidential information.
- **Denial-of-Service attacks** - Attacks that overwhelm a system or network with excessive traffic, making it unavailable to legitimate users.

Main Areas of Cybersecurity

- **Network Security** – Protects computer networks from unauthorized access and cyberattacks.
Example: A company uses a firewall to block hackers from accessing its internal network.
- **Application Security** – Protects software and applications from vulnerabilities and attacks.
Example: An online banking app uses secure coding and regular updates to prevent hacking.
- **Cloud Security** – Protects data, applications, and services stored in cloud platforms.
Example: Google Drive encrypts files to keep users' data secure.
- **Mobile Security** – Protects smartphones, tablets, and mobile apps from cyber threats.
Example: A banking app requires fingerprint or Face ID authentication before access.
- **Endpoint Security** – Protects endpoint devices such as laptops, desktops, and servers from malware and unauthorized access.
Example: A company installs antivirus software on all employee laptops.
- **IoT Security** – Protects Internet of Things (IoT) devices connected to the internet from cyber threats.
Example: Securing smart CCTV cameras with strong passwords and regular firmware updates.
- **Identity and Access Management (IAM)** – Ensures only authorized users can access systems and resources.
Example: A university requires students to log in using their school ID and password with Multi-Factor Authentication (MFA).

Real-World Example: GCash Phishing

Attackers create fake GCash websites that closely resemble the official login page. Users unknowingly enter their credentials, allowing attackers to steal their accounts.

Cybersecurity measures such as:

- Multi-factor authentication (MFA)
- Secure websites (HTTPS)
- Fraud detection systems
- **User awareness campaigns**

Analysis

Cybersecurity focuses specifically on **digital threats**. It includes technologies such as firewalls, antivirus software, intrusion detection systems, and secure authentication methods. However, it is only one component of a broader Information Assurance strategy.

3. What is Information Security?

Information Security (InfoSec) is the practice of protecting information from unauthorized access, disclosure, modification, destruction, or disruption.

It focuses on protecting information regardless of its format:

- Digital files
- Printed documents
- Emails

- Verbal communication
- Cloud storage
- Physical records

The primary goal of Information Security is to maintain the **Confidentiality, Integrity, and Availability (CIA Triad)** of information.

Examples: Hospital - Patient medical records are accessible only to authorized doctors and nurses.

University - Only instructors can modify student grades.

Company - Employee payroll records are restricted to the Human Resources department.

Real-World Example: Healthcare

Hospitals store sensitive information such as:

- Medical history
- Laboratory results
- Prescriptions
- Personal identification

If these records are leaked or altered, patient privacy and safety can be compromised. Information Security ensures that only authorized personnel can access and modify these records.

Analysis

Information Security is not limited to digital systems. A printed contract left unattended on a desk or a confidential conversation overheard in a public place can also result in information security incidents.

4. Difference between Information Security, Assurance and Cybersecurity

Difference Between Information Assurance, Information Security, and Cybersecurity

Information Assurance	Information Security	Cybersecurity
Broadest discipline	Protects information	Protects digital systems
Includes governance, policies, and risk management	Protects information in any form	Focuses on cyber threats
Includes legal and business continuity considerations	Covers both physical and digital information	Primarily protects electronic assets

Example: Imagine a university's online enrollment system.

Cybersecurity

- Blocks hackers from accessing the website.
- Protects the server against malware.
- Uses HTTPS for secure communication.

Information Security

- Ensures only authorized staff can modify student records.
- Protects printed enrollment forms.
- Secures backup files.



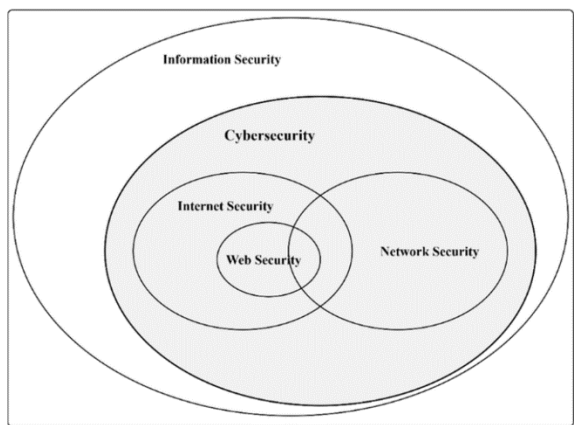
Information Assurance

- Includes all of the above, plus:
 - Risk management
 - Disaster recovery plans
 - Staff training
 - Security policies
 - Legal compliance
 - Regular audits

Analysis

Many students use these terms interchangeably. While they are closely related, Information Assurance is the broadest discipline. It encompasses Information Security, which in turn includes Cybersecurity measures focused on digital environments.

5. Relationship among IA, Information Security, and Cybersecurity



Interpretation

Information Security is the broadest field that protects all forms of information, while Cybersecurity focuses specifically on protecting digital systems and networks. Within Cybersecurity, Internet Security, Network Security, and Web Security are specialized areas that work together to secure online communications, networks, and web applications.

Relationship between information security, cybersecurity, Internet security, network security, and web security, derived from von Solms and von Solms [24] and BSI [21,22].

6. Goals of Information Assurance

The main goals of Information Assurance include:

- Protect information from unauthorized access.
- Maintain data accuracy and integrity.
- Ensure information is available when needed.
- Verify user identities before granting access.
- Ensure accountability for user actions.
- Support business continuity during disruptions.
- Comply with legal and regulatory requirements.

Real-World Example: Online Banking - A customer accesses an online banking system.

Information Assurance ensures:

- The customer logs in using valid credentials.
- Transactions are recorded accurately.
- Banking services remain available during peak usage.
- Fraudulent activities are detected and investigated.
- Transaction records can be audited if disputes arise.

7. Importance of Information Assurance

Information Assurance is essential because organizations depend on information to operate effectively.

Without IA, organizations may experience:

- Financial losses
- Legal penalties
- Loss of customer trust
- Operational disruptions
- Data breaches
- Identity theft
- Reputational damage

Philippine Case Study: PhilHealth Ransomware Attack (2023)

In September 2023, PhilHealth was hit by the Medusa ransomware group. The attack disrupted operations and raised concerns about the security of members' personal information.

Lessons Learned

- Regular backups are essential.
- Incident response plans should be well established.
- Security awareness training helps reduce human error.
- Continuous system monitoring is necessary to detect threats early.

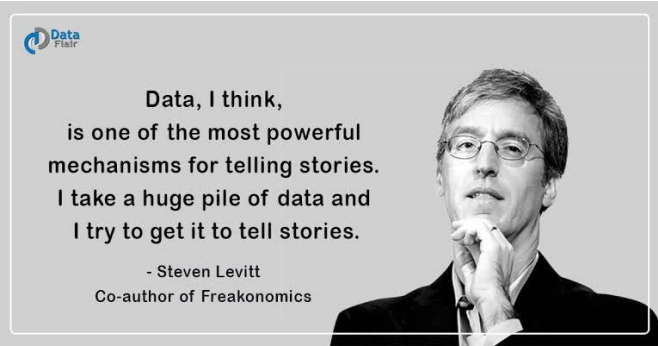
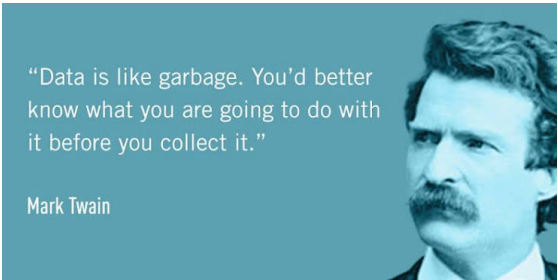
This incident highlights that Information Assurance involves not only preventing attacks but also ensuring organizations can respond to and recover from them.

8. Information as an Organizational Asset

Information is one of an organization's most valuable resources because it supports decision-making, operations, customer service, and regulatory compliance.

Examples of Information Assets

- Customer databases
- Employee records
- Financial statements
- Payroll information
- Intellectual property
- Research data
- Student records
- Medical records
- Government documents
- Source code
- System configurations
- Cloud databases



Real-World Example: E-Commerce Company -
An online retailer stores:

- Customer names
- Delivery addresses
- Payment information
- Purchase history

If these data are stolen, the company may suffer financial losses, legal consequences, and reputational damage. Protecting these assets is therefore a critical business priority.

Analysis

In today's digital economy, data is often described as the **"new oil"** because it drives innovation, strategic decisions, and competitive advantage. However, unlike physical assets, information can be copied, altered, or transmitted instantly. This makes effective information assurance essential for organizations of all sizes.

Recommended Learning Resources

Books

- Whitman, M. E., & Mattord, H. J. *Principles of Information Security* (latest edition)
- Ciampa, M. *CompTIA Security+ Guide to Network Security Fundamentals*
- Stallings, W. *Network Security Essentials*

Official References

- NIST Cybersecurity Framework: <https://www.nist.gov/cyberframework>
- CISA Cybersecurity Resources: <https://www.cisa.gov>
- OWASP Foundation: <https://owasp.org>
- Microsoft Learn – Security Fundamentals: <https://learn.microsoft.com/training/>
- Cisco Networking Academy – Introduction to Cybersecurity: <https://www.netacad.com/courses/cybersecurity/introduction-cybersecurity>
- National Privacy Commission (Philippines): <https://privacy.gov.ph>